

Object-Oriented Programming in Common Lisp: A Programmer's Guide to CLOS

Sonya E. Keene

Chapter 1 Introduction to the CLOS Model

Chapter 2 Elements of CLOS Programs

- 2.1 Classes and Instances
- 2.2 Slots
- 2.3 Superclasses
- 2.4 Generic Functions
- 2.5 Methods
- 2.6 Method Rules
- 2.7 The Controller of Inheritance
- 2.8 Summary of the CLOS Model
- 2.9 How CLOS Extends COMMON LISP

Chapter 3 Developing a Simple CLOS Program: Locks

- 3.1 Overview of Locking
- 3.2 Defining the Kinds of Objects—Classes
- 3.3 Creating New Objects—Instances
- 3.4 Defining the Interface—Generic Functions
- 3.5 Defining the Implementation—Methods
- 3.6 Specializing the Behavior of Locks
- 3.7 Analyzing the Inheritance of Locks
- 3.8 Extending the Locking Program
- 3.9 How Client Programs Use Locks
- 3.10 Reviewing the Lock Classes
- 3.11 The External and Internal Perspectives
- 3.12 Guidelines on Designing Protocols

Chapter 4 Programming with Methods

- 4.1 Implementation Choices: Methods versus Slots
- 4.2 Programming with Accessors
- 4.3 Multi-Methods
- 4.4 Methods for COMMON LISP Types
- 4.5 Methods for Individual LISP Objects
- 4.6 Summary of Method Inheritance

Chapter 5 Controlling the Generic Dispatch

- 5.1 The Core Framework
- 5.2 Declarative and Imperative Techniques
- 5.3 Around-Methods
- 5.4 Calling a Shadowed Primary Method
- 5.5 Using a Different Method Combination Type
- 5.6 Built-in Method Combination Types
- 5.7 Defining a New Method Combination Type
- 5.8 Guidelines on Controlling the Generic Dispatch
- 5.9 Summary of the Generic Dispatch Procedure
- 5.10 Summary of the Standard Method Combination Type

Chapter 6 Class Inheritance

- 6.1 Inheritance from Default Classes
- 6.2 The Class Precedence List
- 6.3 Guidelines on Designing Class Organizations
- 6.4 Inheritance of Slots and Slot Options
- 6.5 Guidelines on Using Inheritance of Slot Options

Chapter 7 Defining CLOS Elements

- 7.1 Order of Defining CLOS Elements
- 7.2 Congruent Lambda-Lists
- 7.3 LISP Object Representing CLOS Elements
- 7.4 Mapping Between Names and Objects
- 7.5 Removing Generic Functions and Methods

Chapter 8 Redefining CLOS Elements

- 8.1 Redefining Classes
- 8.2 Redefining Methods and Generic Functions
- 8.3 Example of Redefining CLOS Elements
- 8.4 Changing the Class of an Instance

Chapter 9 Creating and Initializing Instances

- 9.1 Arguments to `make-instance`
- 9.2 Summary of What `make-instance` Does
- 9.3 Controlling Initialization with `defclass` Options
- 9.4 Controlling Initialization with Methods
- 9.5 Initialization Arguments
- 9.6 Constructors

Chapter 10 A Procedural Definition: Initialization

- 10.1 Examples of Procedural Definitions
- 10.2 Isolating Work Shared Among Procedures
- 10.3 Filling Unbound Slots with `initforms`
- 10.4 Performing Initialization by `initargs`
- 10.5 Specializing a Portion of the Procedure
- 10.6 Declaring `initarg` Names as Valid

Chapter 11 Developing an Advanced CLOS Program: Streams

- 11.1 Overview of Streams
- 11.2 Design of Our Stream Foundation
- 11.3 Experimenting with the Stream Example
- 11.4 Directional Streams
- 11.5 Tape Streams
- 11.6 Disk Streams
- 11.7 Character Streams
- 11.8 Byte Streams
- 11.9 Instantiable Streams
- 11.10 A Procedural Definition for Creating Streams
- 11.11 Summary of Techniques Used in Streams

11.12 Documenting the Implementation of Streams

Chapter 12 Highlights of CLOS

12.1 Designing a Standard Object-Oriented Paradigm

12.2 Summary of the Programmer Interface

12.3 Exploring Alternative Paradigms

12.4 A Nongoal: Automatic Protocol Support

12.5 The CLOS Metaobject Protocol

Appendix A Glossary of CLOS Terminology

Appendix B Syntax of CLOS Operators

Appendix C CLOS Operators Not Documented in This Book

Bibliography

Index